

Exercise 1 – LTspice

Use the file on Brightspace: **Week6_Ex1.asc**. It contains the setup to simulate the noise of resistive-loaded CS amplifier.

- A. Run the noise analysis. Plot the input-referred and output-referred noise.
 - Calculate the noise components using the small-signal models. Use the small-signal parameters from the `.op` analysis (also compare those to your expectations).
 - Compare calculations and simulation. Which noise source is dominating at each frequency range?
- B. Add a cascode with the same size as M1 and bias its gate so that M1 has a margin of 50 mV in VDS from the triode region. To keep both transistors in saturation, reduce the load resistor to 5 k Ω (Why is this required?)
 - Repeat the noise analysis and compare results to your expectations.
 - What changed w.r.t. to simulations in point A?
- C. Reduce the cascode gate voltage by 200 mV and execute the noise analysis again. What changes do you see?
 - Repeat the noise analysis and compare results to your expectations.
 - What changed w.r.t. to simulations in point A?

Exercise 2 – LTspice

Use the file on Brightspace: **Week6_Ex2.asc**. It contains the setup to simulate the noise of a telescopic amplifier. Note that all sizes are parametrized to help scaling. The goal of this exercise is to identify dominating noise sources and see how we can change the design to improve the noise behaviour.

A. Find the dominating noise sources

- Perform a noise simulation, the noise of which 4 transistors dominate?
- Show a plot of the input referred thermal noise and flicker noise voltage spectral density of these 4 transistors.
Hint: To plot the flicker noise or thermal noise only, right-click on the plot and look under 'add traces'.
 $V(mX.1overf)$ is flicker noise
 $V(mX.ds)$ is thermal noise
- Also give the integrated input referred RMS noise voltage of all 4 transistors.
Hint: The integrated noise can be found using 'cntrl+left-click' on the trace name in the plot window.

Exercise 2.B – LTspice

- What is the noise contribution of the NMOS which provides the tail current (M10) and the noise contribution of the biasing network? Explain why.

Exercise 2.C – LTspice

- For low frequencies, the Flicker noise dominates. We want to reduce the noise at these low frequencies.
- Scale the input transistors (M3 and M15) such that the input referred flicker noise becomes lower than $3 \mu V_{RMS}$, while maintaining a gain larger than 63 dB and without increasing the power consumption.
- *Hint: Use parameters at top of schematic*

Exercise 2.D – LTspice

- At this point, the thermal noise dominates over the flicker noise.
- Reduce the thermal noise of the input transistors by a factor 3. You are allowed to increase the power consumption, but keep the current sources at the same values.
Furthermore the gain should not (significantly) be affected.

Exercise 2.E – LTspice

- Up until this point we used a perfectly symmetric amplifier. In reality, the amplifier will not be completely symmetric.
- Introduce a DC differential bias at the input of 20 mV. What happens to the noise from the tail current and the connected current mirror (M10 and M4)? Why?
- Explain the difference between the noise from M10 and M4.